ASSIGNMENT 01

CLOUD COMPUTING

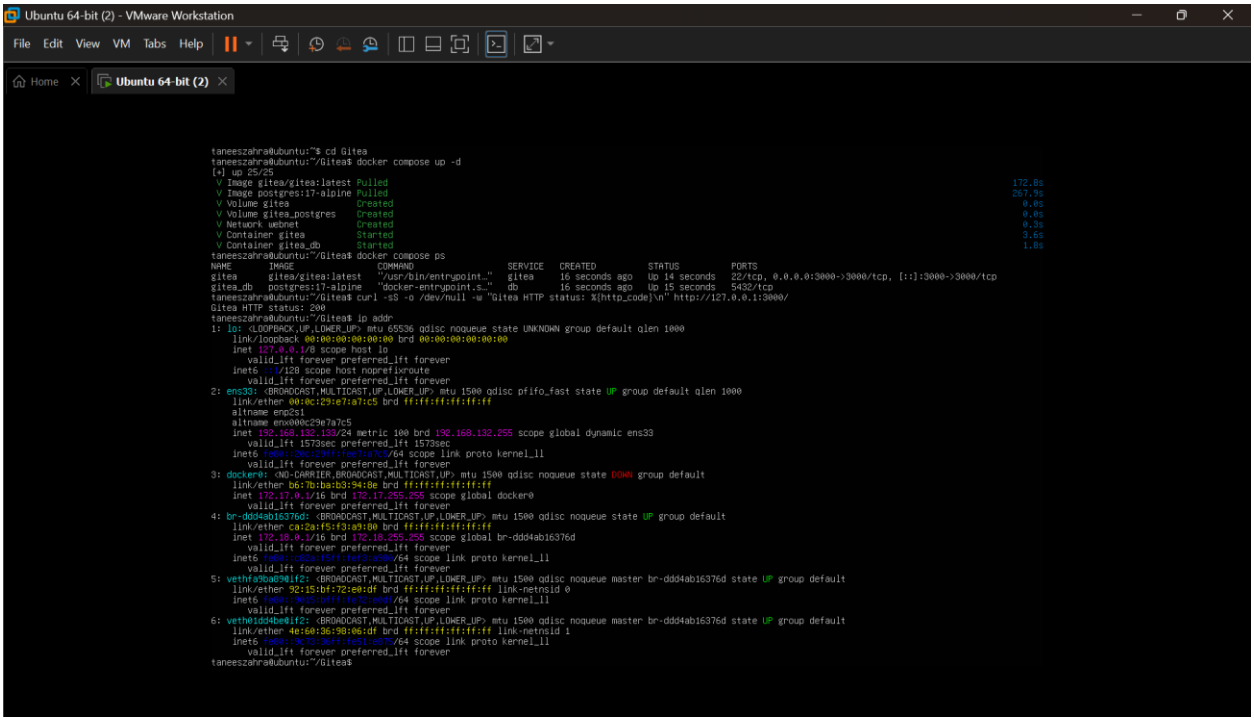
Git, Gitea, GitHub, Git LFS, and GitHub Pages

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

1.1 Objective

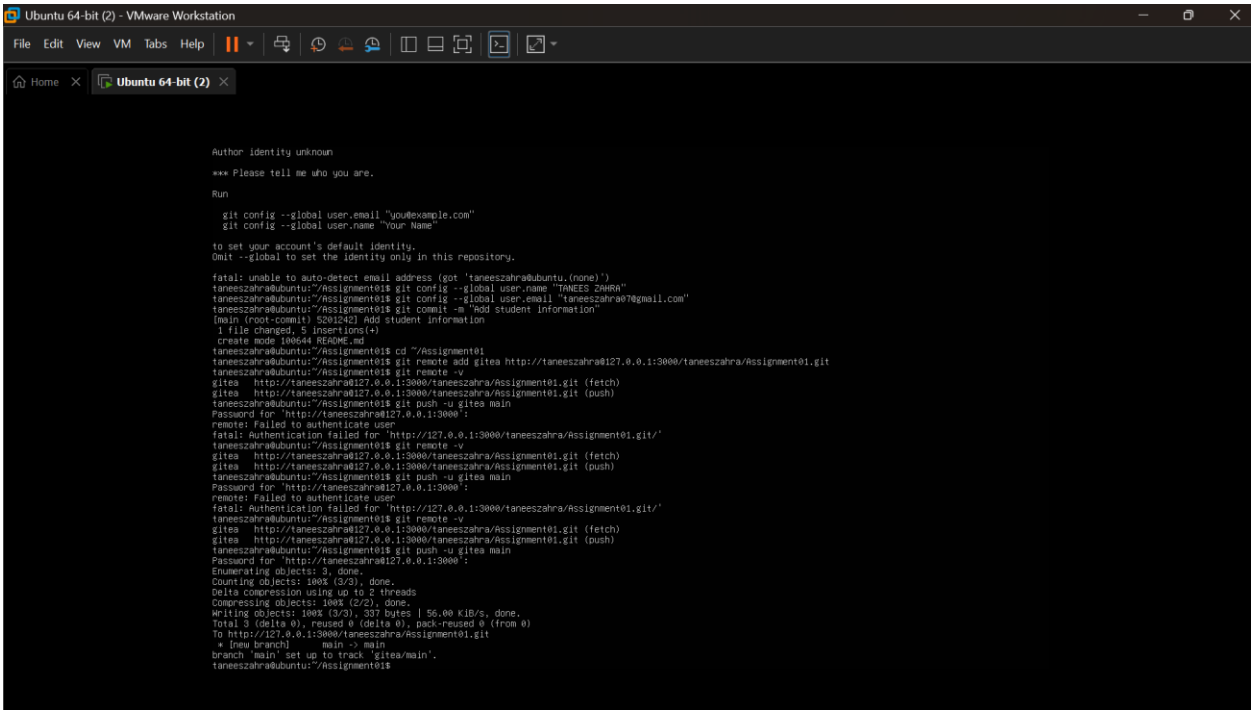
Install and run a local Gitea server on Ubuntu Server, create a user profile, and initialize the `Assignment01` repository.

Gitea Web Interface Running:



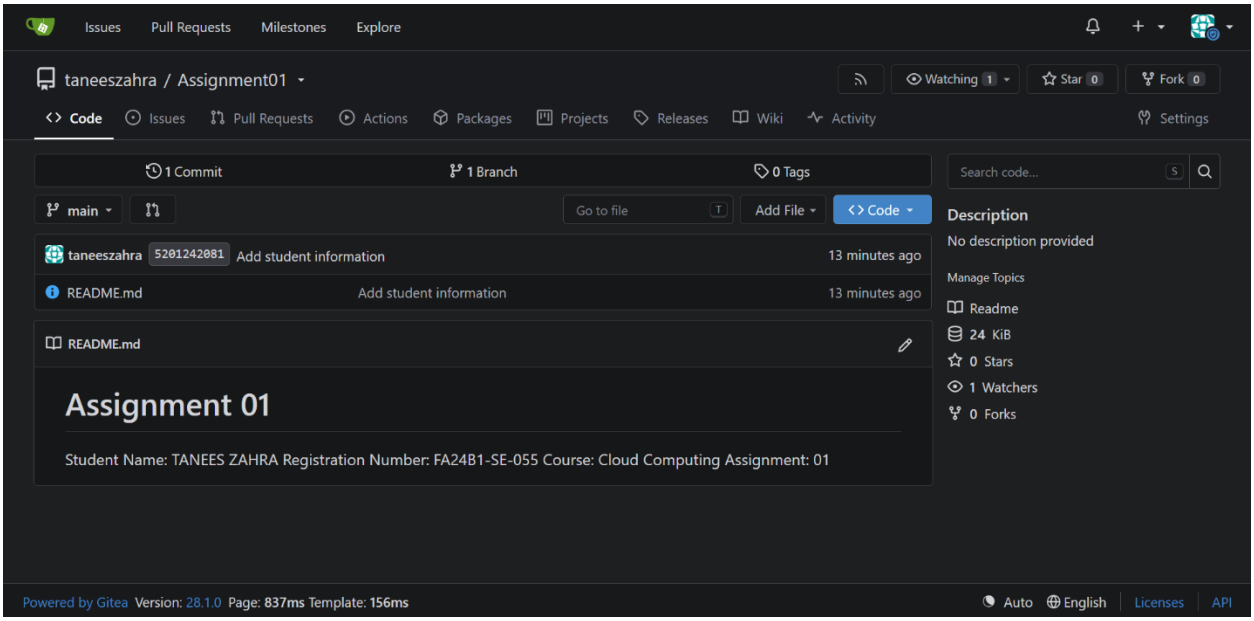
task1_gitea_running.png

Gitea Repository Created:



task1_gitea_repository.png

Initial Push to Gitea Remote:



task1_gitea_push.png

Task 2: Multi-Remote Repository Setup (Gitea & GitHub)

2.1 Objective

Configure dual remotes (`gitea` and `github`) within the local Git repository and synchronize commits across both platforms using Personal Access Tokens (PAT).

Terminal Verification of Git Remotes:

```
taneeszahra@ubuntu:~/Assignment01$ cd ~/Assignment01
taneeszahra@ubuntu:~/Assignment01$ git remote add github https://github.com/taneeszahra/Assignment01.git
taneeszahra@ubuntu:~/Assignment01$ git remote -v
gitea   http://taneeszahra@127.0.0.1:3000/taneeszahra/Assignment01.git (fetch)
gitea   http://taneeszahra@127.0.0.1:3000/taneeszahra/Assignment01.git (push)
github  https://github.com/taneeszahra/Assignment01.git (fetch)
github  https://github.com/taneeszahra/Assignment01.git (push)
taneeszahra@ubuntu:~/Assignment01$
```

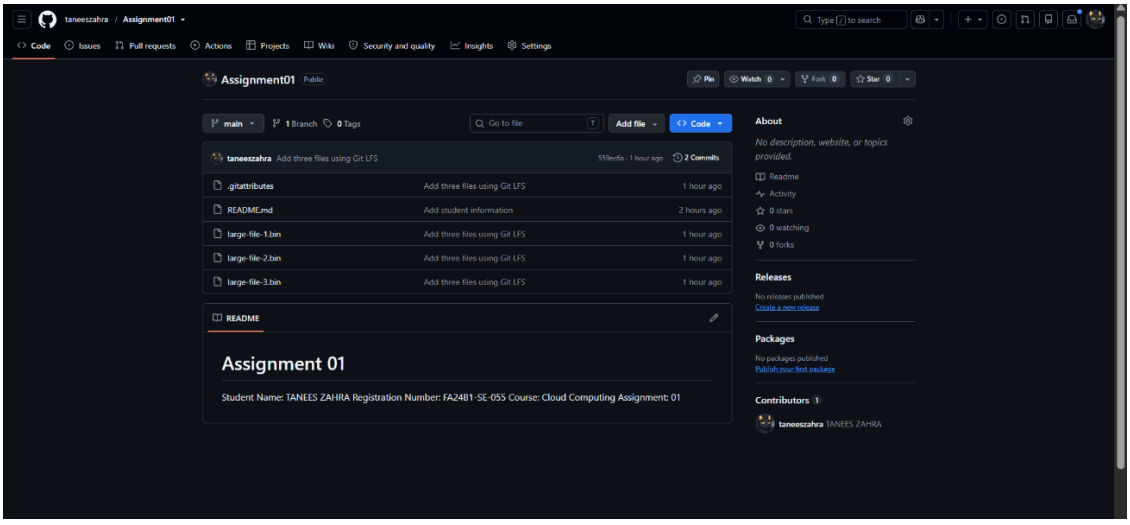
task2_remotes.png

Terminal Push to GitHub Remote:

```
taneeszahra@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': taneeszahra
Password for 'https://taneeszahra@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 337 bytes | 18.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/taneeszahra/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
taneeszahra@ubuntu:~/Assignment01$ _
```

task2_github_push.png

GitHub Web Interface Showing Synchronized Repository:



task2_github_repository.png

Task 3: Git Large File Storage (Git LFS) Integration

3.1 Objective

Install Git LFS, configure tracking for `.bin` binary files, commit large placeholder files, and push to remote repositories.

Git LFS Installation and Tracking Setup:

```
branch 'main' set up to track 'github/main'.
taneeszahra@ubuntu:~/Assignment01$ cd ~/Assignment01
taneeszahra@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
taneeszahra@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
taneeszahra@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
taneeszahra@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
taneeszahra@ubuntu:~/Assignment01$ _
```

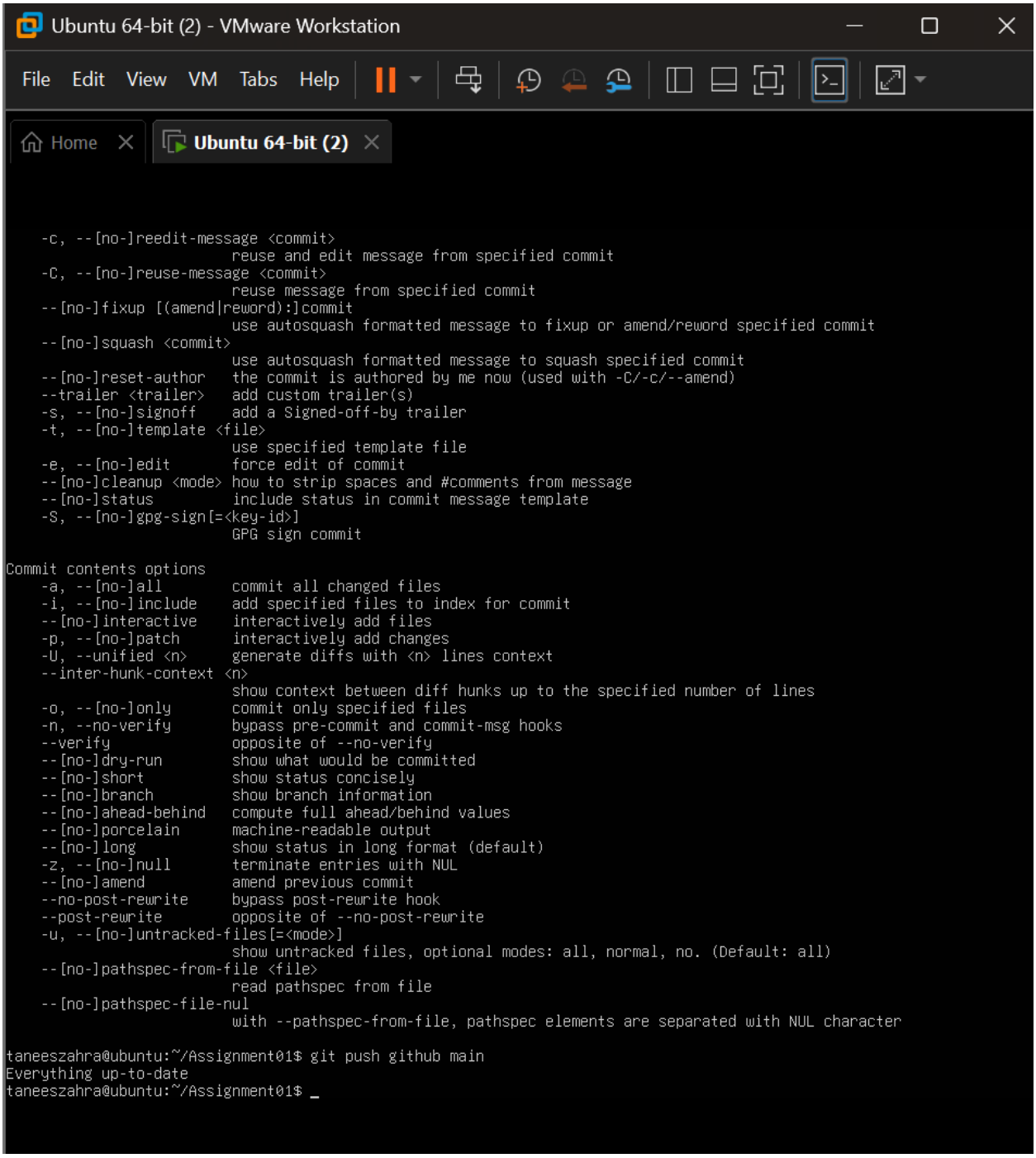
task3_lfs_setup.png

Git LFS File Tracking Status (git lfs ls-files):

```
rm 'large-files-3.bin'
taneeszahra@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
taneeszahra@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
taneeszahra@ubuntu:~/Assignment01$ truncate -s 1-3M large-file-3.bin
truncate: Invalid number: '1-3M'
taneeszahra@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
taneeszahra@ubuntu:~/Assignment01$ ls -lh large-file-*.bin
-rw-rw-r-- 1 taneeszahra taneeszahra 101M Oct 10 04:46 large-file-1.bin
-rw-rw-r-- 1 taneeszahra taneeszahra 102M Oct 10 04:47 large-file-2.bin
-rw-rw-r-- 1 taneeszahra taneeszahra 103M Oct 10 04:47 large-file-3.bin
taneeszahra@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
taneeszahra@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
taneeszahra@ubuntu:~/Assignment01$
```

task3_lfs_files.png

Git LFS Object Push Output:



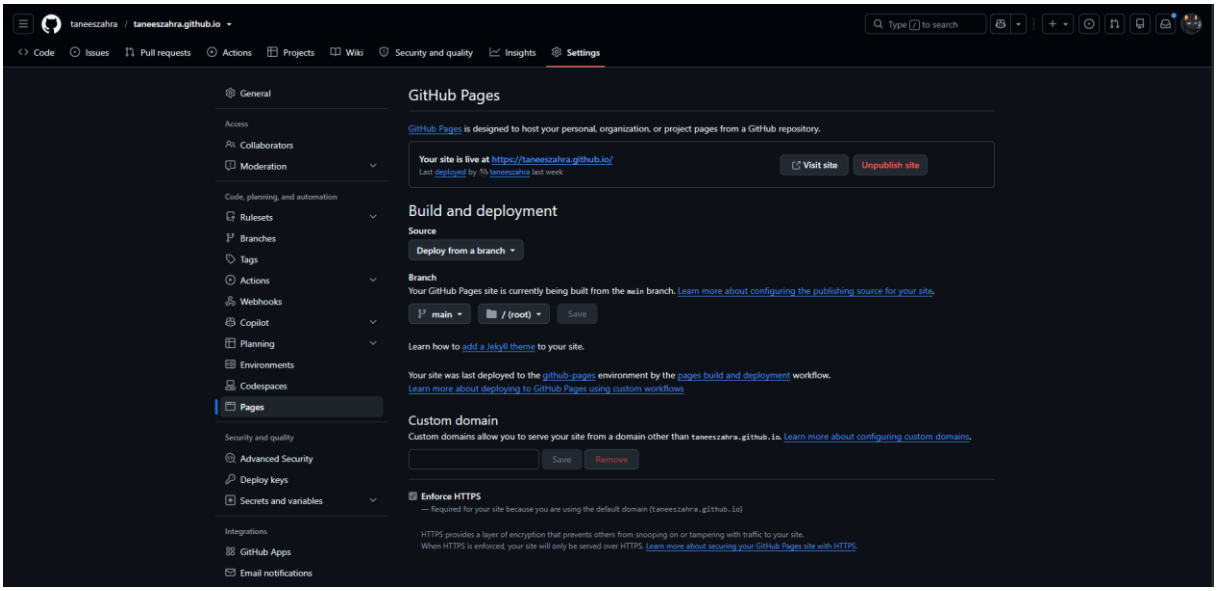
task3_lfs_push.png

Task 4: Responsive Portfolio Deployment via GitHub Pages

4.1 Objective

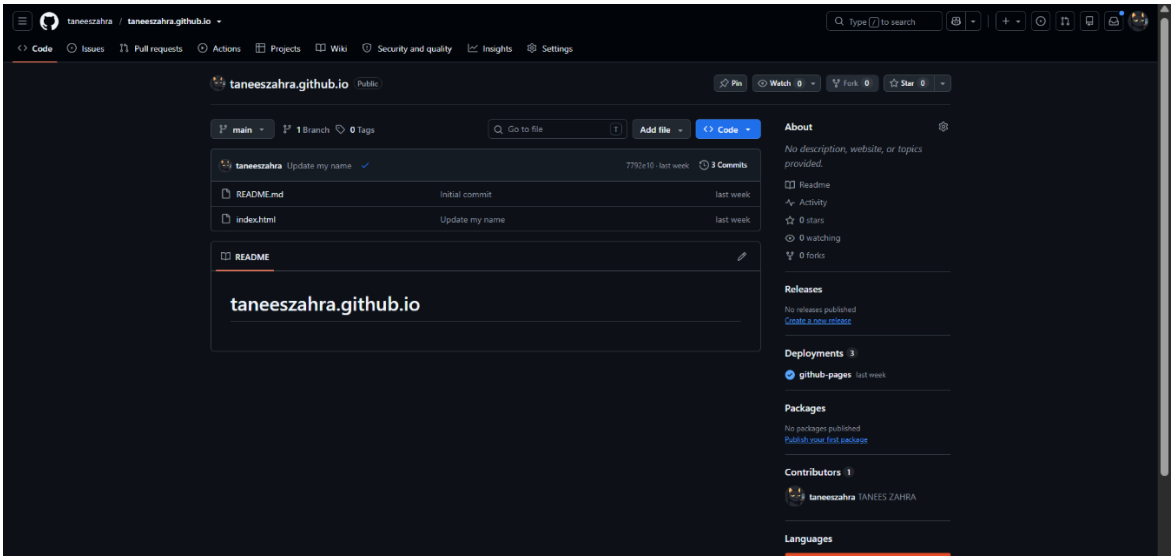
Design and deploy a modern responsive developer portfolio site tailored for a Software Engineer profile using HTML5, CSS3 Glassmorphism UI styling, and hosting via GitHub Pages.

GitHub Pages Configuration Page:



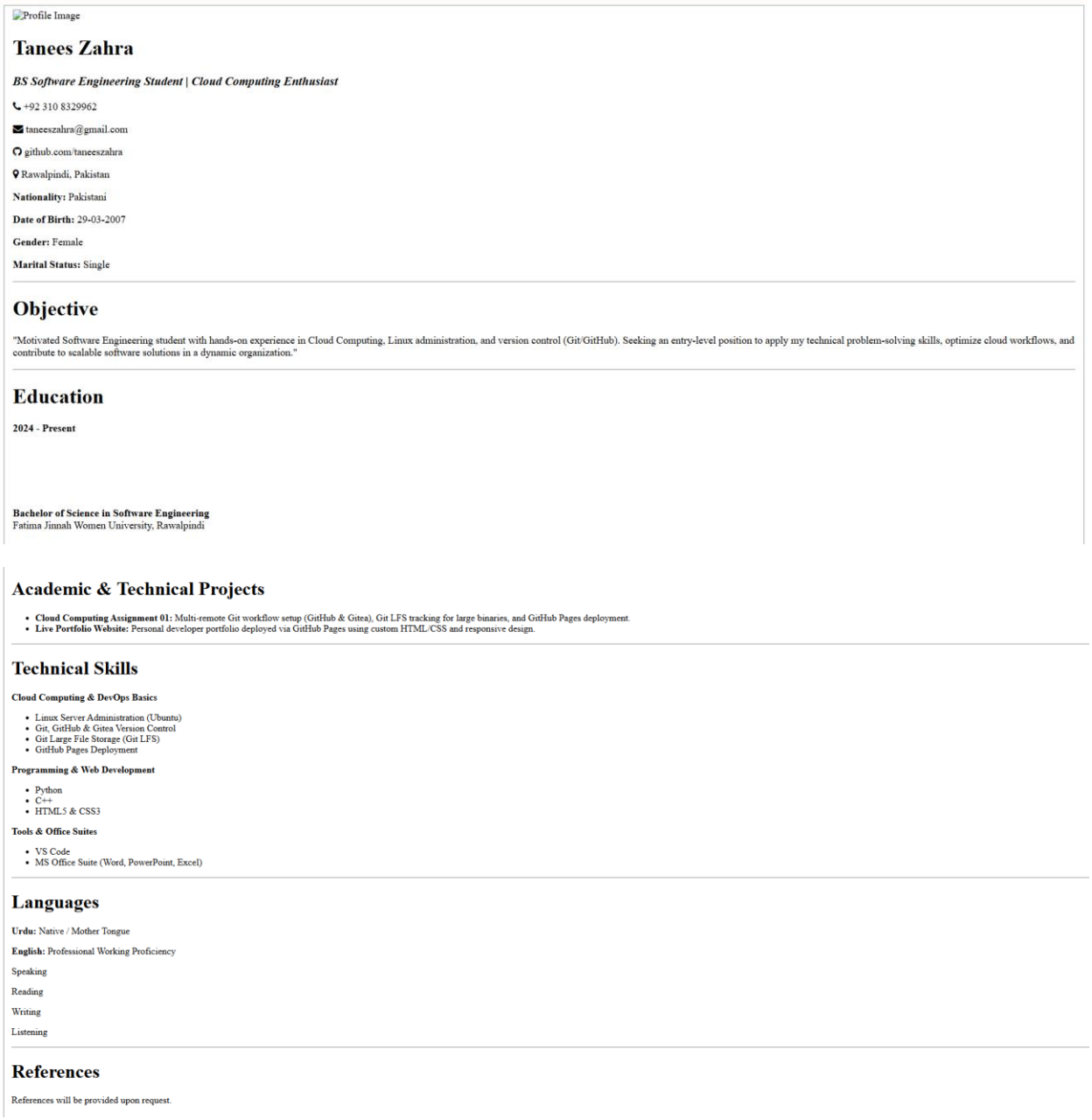
task4_pages_deployment.png

Portfolio Repository Structure on GitHub:



task4_pages_repository.png

Live Deployed Portfolio Web Page View:



Profile Image

Tanees Zahra

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Rawalpindi, Pakistan

Nationality: Pakistani

Date of Birth: 29-03-2007

Gender: Female

Marital Status: Single

Objective

"Motivated Software Engineering student with hands-on experience in Cloud Computing, Linux administration, and version control (Git/GitHub). Seeking an entry-level position to apply my technical problem-solving skills, optimize cloud workflows, and contribute to scalable software solutions in a dynamic organization."

Education

2024 - Present

Bachelor of Science in Software Engineering
Fatima Jinnah Women University, Rawalpindi

Academic & Technical Projects

- **Cloud Computing Assignment 01:** Multi-remote Git workflow setup (GitHub & Gitea), Git LFS tracking for large binaries, and GitHub Pages deployment.
- **Live Portfolio Website:** Personal developer portfolio deployed via GitHub Pages using custom HTML/CSS and responsive design.

Technical Skills

Cloud Computing & DevOps Basics

- Linux Server Administration (Ubuntu)
- Git, GitHub & Gitea Version Control
- Git Large File Storage (Git LFS)
- GitHub Pages Deployment

Programming & Web Development

- Python
- C++
- HTML5 & CSS3

Tools & Office Suites

- VS Code
- MS Office Suite (Word, PowerPoint, Excel)

Languages

Urdu: Native / Mother Tongue

English: Professional Working Proficiency

Speaking

Reading

Writing

Listening

References

References will be provided upon request.

task4_portfolio_live.png

Conclusion

All requirements for Cloud Computing Assignment #1 were completed successfully. The local Gitea instance and GitHub remote synchronizations are functioning, Git LFS correctly handles binary pointers without overloading history, and the personal developer portfolio is live online.